



DoubleML

DoubleML - Introduction and Workflow

DoubleML Workshop

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Economic AI

Welcome to our DoubleML Training!

Meet the DoubleML Team



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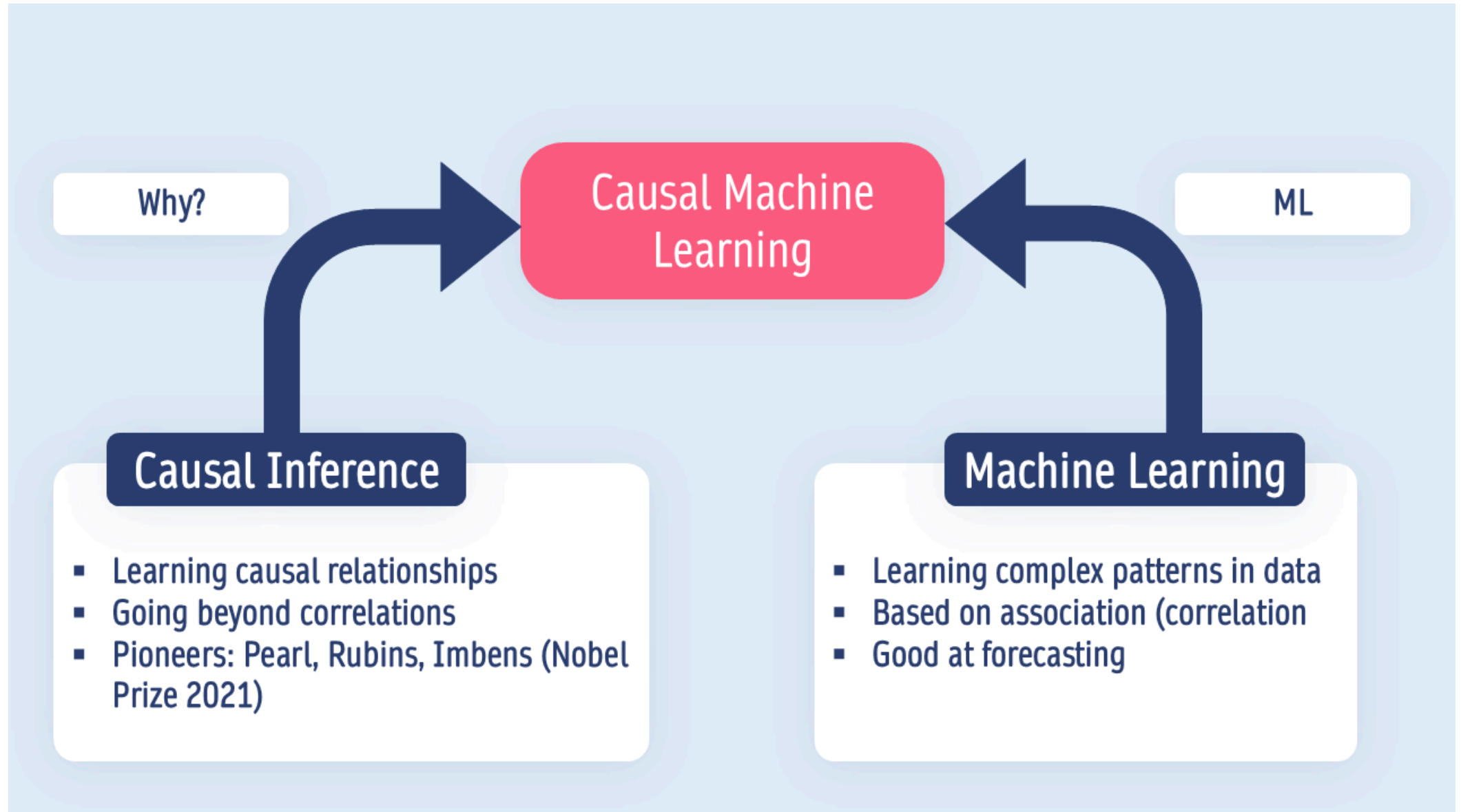
Jan Teichert-Kluge



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Basics: Causality and Causal ML

Basics: Causal ML



Basics: Causal ML

Causal Modelling

- Look into the black box and learn about the underlying mechanism
- Example: Association between lung cancer and wearing a lighter
 - Obviously, no causal relationship
 - Nevertheless, prediction might work!

→ Causal understanding is key for real “Intelligence”



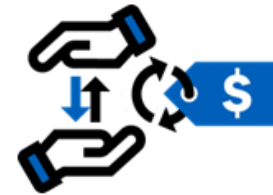
Resource Allocation



A/B Testing



Uplift Modelling



Dynamic Pricing



Clinical Trials



Production Optimization

Basics: Predictive vs. Causal Modeling

Predictive Modeling

How can we build a good prediction rule, $f(X)$, that uses features X to predict Y ?

Example: **Customer Churn**

“How well can we predict whether customers churn?”

“Which of these variables are good predictors for churn (explainability)?”

Causal Modeling

What is the causal effect of a treatment D on an outcome Y ?

“Why do customer churn?”

“How can we retain customers?”

Causal ML: How can we use state-of-the art ML methods for causal inference?

Causal Inference: A Brief Introduction

Introduction to Causal Inference

How to assess causality with data?

- **Approach**

1. Define *causal* parameters of interest
2. State *necessary assumptions* for
 - **Identification** and
 - **Valid estimation**

- **Helpful *tools* in causal inference**

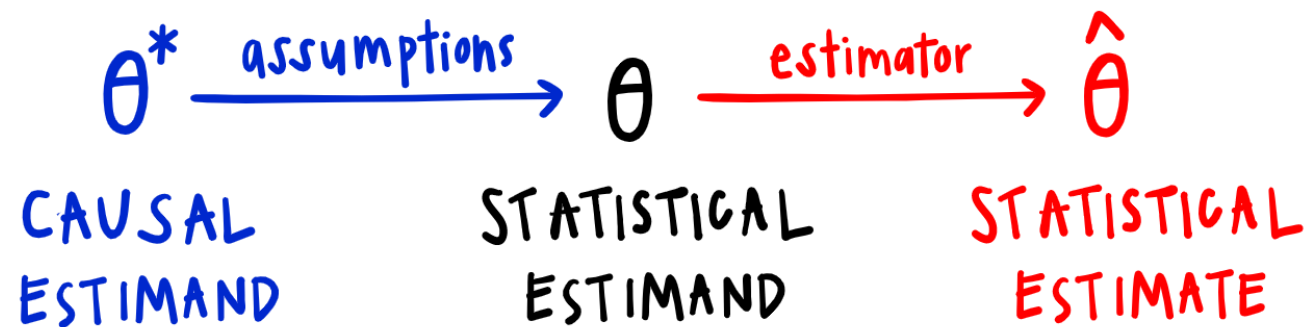
- **Potential Outcomes** (PO) framework
- **Directed Acyclic Graphs** (DAGs)

General question

What is the causal effect of **treatment** D on **outcome** Y ?

Introduction to Causal Inference

IDENTIFICATION vs. ESTIMATION



What hypothetical quantity, or parameter, are we interested in?

Can we write this causal quantity in terms of observable data?

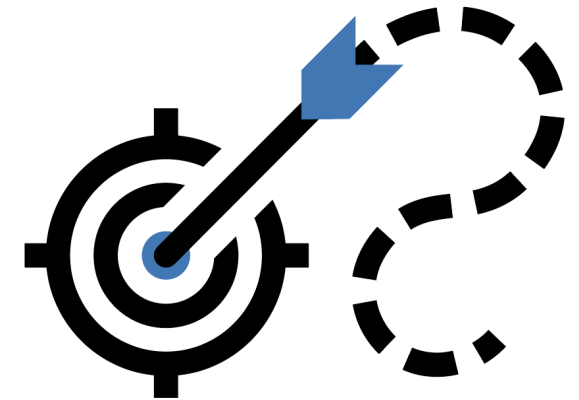
What algorithm will best approximate this statistical quantity?

Introduction to Causal Inference

Example: Uplift Modeling

Key questions:

1. What is the (causal) effect of an **email campaign (coupon)** ($= D$) on **product sales** (conversion) ($= Y$)?
2. How to optimally target **coupons** ($= D$) towards newsletter subscribers?



Definition: Causal Effects

- Binary treatment D

$$D = \begin{cases} 1, & \text{if treated (= with coupon)} \\ 0, & \text{if not treated (= without coupon)} \end{cases}$$

Potential outcome framework

- Define the **potential outcomes** as
 - $Y(1)$: Conversion if **she would receive a discount**
 - $Y(0)$: Conversion if **she would not receive a discount**
- Individual causal effect (ICE)

$$\Delta = Y(1) - Y(0)$$

Fundamental Problem of Causal Inference

Individual causal effects cannot be identified, in general

- We only observe one of the potential outcomes $\Rightarrow$ *Factual*
- The other outcome is unobserved $\Rightarrow$ *Counterfactual*

What can we do about this?

- Other causal quantities, like average individual treatment effect (ATE) might be identified
- Additional assumptions, e.g., selection-on-observables assumption, might be required
- Use data and estimate the causal parameters

Estimation of Causal Effects

- A/B test, **R**andomized **C**ontrol **T**rial (RCT), experiment
- Often considered as “Gold Standard” for causal inference
- Under certain assumptions, we can consistently estimate the **A**verage **T**reatment **E**ffect (ATE)¹ of treatment D on outcome Y

$$ATE = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)]$$

Basic idea:

- Use sample-based estimates $\mathbb{E}[Y | D = 1]$ and $\mathbb{E}[Y | D = 0]$ for $\mathbb{E}[Y(1)]$ and $\mathbb{E}[Y(0)]$, respectively

Estimation of Causal Effects

Why does it work?

In an RCT, the allocation of the treatment to individuals is assigned randomly

⇒ Treatment assignment is **independent** of **potential outcomes**, $D \perp\!\!\!\perp Y(0), Y(1)$

- Individuals *do not self-select into the treatment*, i.e., they cannot pick the value of the treatment that is best for them (in terms of their potential outcomes)

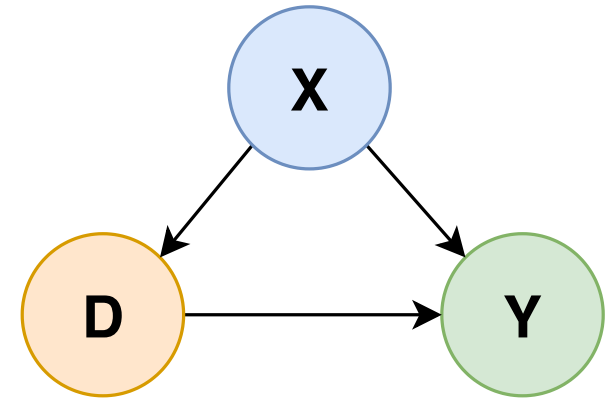
But how to evaluate causal effects without randomization?

Causal Effects from Observational Data

Conditional Exogeneity (Selection-on-Observables)

- Without randomization, treatment and potential outcomes may be related (*confounding*)
- **Key assumption:** Conditional on observed covariates X , treatment is “as good as random”

$$D \perp\!\!\!\perp Y(0), Y(1) \mid X$$



Intuition: If we compare individuals with the *same* characteristics X , any difference in outcomes can be attributed to the treatment (“apples to apples”).

Estimation approaches

- **Regression adjustment:** Model $\mathbb{E}[Y \mid D, X]$ and average over X
- **Reweighting (IPW):** Weight observations by inverse propensity scores $P(D = 1 \mid X)$
- **Doubly robust methods:** Combine both approaches

Introduction to Double Machine Learning

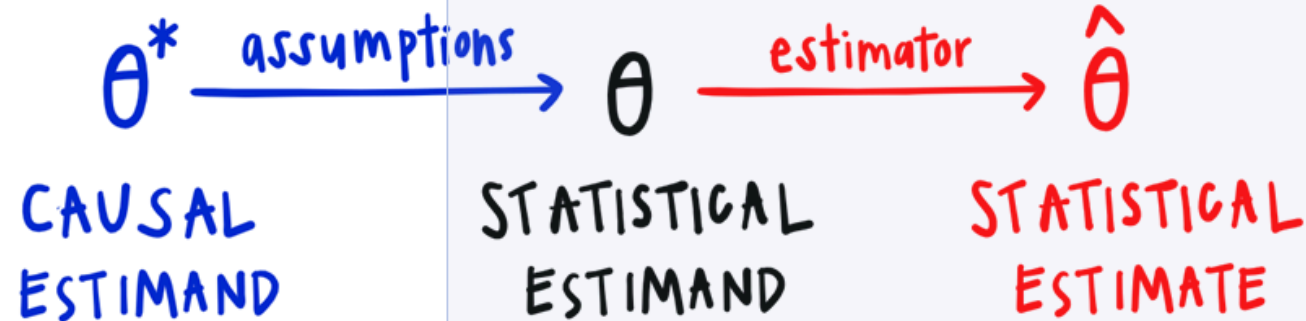
Introduction to Double Machine Learning

Why ML for causal analysis?

- Machine Learning (ML) algorithms are powerful tools for predictions
- Use Case continued: Uplift modeling
 - Experiment, A/B test: Improve precision/efficiency/power
 - Account for confounding in observational studies
 - **High-dimensional** vector of covariates or **complex relationship** of Y , D and X

Introduction to Double Machine Learning

IDENTIFICATION vs. ESTIMATION



What hypothetical quantity, or parameter, are we interested in?

Can we write this causal quantity in terms of observable data?

What algorithm will best approximate this statistical quantity?

Introduction to Double Machine Learning

Why ML for causal analysis?

Prediction – Machine Learning

Powerful methods in high-dimensional and non-linear settings, e.g.,

- Scikit-learn
- Lasso (Lasso, LassoCV)
- Random Forest (RandomForest)
- Boosted Trees (LightGBM, xboost, CatBoost)
- ...

Inference – Econometrics & Statistics

Statistical framework for estimation of causal effects, i.e.,

- Structural equation models,
- Identification,
- Asymptotic properties,
- Hypothesis tests, confidence intervals,
- ...

```
graph TD; A[Prediction – Machine Learning] --> C[Causal / Double Machine Learning]; B[Inference – Econometrics & Statistics] --> C;
```

Causal / Double Machine Learning

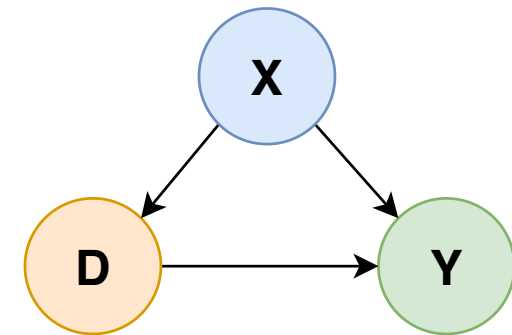
The Challenge: Why Not Just “Plug In” ML Predictions?

Naive Approach

1. Use ML to estimate **nuisance functions** (e.g., $\mathbb{E}[Y|X]$, $P(D = 1|X)$)
2. Plug in predictions to estimate treatment effect θ_0

The Problem

- ML methods use **regularization** (e.g., penalization, early stopping)
- Regularization introduces **bias** in nuisance estimates
- This bias **propagates** to the causal estimate $\hat{\theta}_0$



Intuition: ML optimizes for *prediction accuracy*, not for *unbiased causal estimation*

We need a framework that “immunizes” causal estimates against ML’s regularization bias

Ingredient 1: Why Orthogonality?

Small errors in nuisance estimates can cause large bias

The setup:

- We want to estimate causal parameter θ_0
- We also need to estimate **nuisance functions** η_0 (e.g., conditional expectations, propensity scores)
- ML gives us $\hat{\eta}$ with some estimation error

The problem with naive approaches:

- Estimation error in $\hat{\eta}$ creates **first-order bias** in $\hat{\theta}_0$
- Even small ML errors $\rightarrow$ substantial bias in causal estimate
- Standard inference (confidence intervals, p-values) becomes **invalid**

Ingredient 1: Why Orthogonality?

Orthogonal scores make estimation insensitive to nuisance errors

Key idea:

- Construct **score functions** where errors in $\hat{\eta}$ have only **second-order** effects on $\hat{\theta}_0$
- Small errors in nuisance $\times$ small errors = negligible impact

What this achieves:

- ML regularization bias in $\hat{\eta}$ doesn't "infect" $\hat{\theta}_0$
- We can use flexible ML methods and still get valid inference
- Confidence intervals have correct coverage

Geometric intuition:

- Think of θ and η as directions
- **Orthogonal** = perpendicular
- Moving slightly in the η direction doesn't change your position in the θ direction

Good News: DoubleML automatically uses orthogonal scores — you don't need to derive them!

Ingredient 2: High-Quality ML Estimation

The nuisance functions must be estimated “well enough”

Why does ML quality matter?

- Orthogonality reduces bias from nuisance errors
- But errors can't be arbitrarily large
- Product of ML errors must shrink fast enough

Practical guidance:

- Use flexible learners appropriate for your data
- Perform hyperparameter tuning
- Evaluate prediction quality with out-of-sample metrics

What this allows:

- ML methods can converge slower than classical parametric rates
- Enables use of Random Forests, Gradient Boosting, Neural Networks, etc.

Rule of Thumb: If your ML predictions are poor, your causal estimates will suffer!

Ingredient 3: Sample Splitting (Cross-Fitting)

Preventing overfitting from contaminating causal estimates:

- ML models can overfit the training data
- Using same data for ML fitting and causal estimation creates bias
- Overfitting errors become correlated with estimation errors

Solution: Cross-fitting

1. Split data into K folds
2. For each fold: train ML on other folds, predict on held-out fold
3. Combine all out-of-sample predictions for causal estimation

Benefits:

- Predictions are always out-of-sample
- All data used for both training and estimation
- More efficient than simple train/test split

Main Result: Double Machine Learning

The Key Ingredients of DML

1. Neyman orthogonality
2. High-quality ML estimation
3. Sample splitting

Main result in Chernozhukov et al. (2018)

There exist regularity conditions, such that the DML estimator $\tilde{\theta}_0$ concentrates in a $1/\sqrt{N}$ -neighborhood of θ_0 and the sampling error is approximately

$$\sqrt{N}(\tilde{\theta}_0 - \theta_0) \sim N(0, \sigma^2).$$

Getting Started with DoubleML

Implementation and Theoretical Framework of DML

Key ingredients

- **Orthogonal Score**
 - Object-oriented implementation
 - Exploit common structure being centered around a (linear) score function $\psi(\cdot)$
- **High-quality ML**
 - State-of-the-art ML prediction and tuning methods
 - Provided by `scikit-learn` and `sckit-learn`-like learners
- **Sample Splitting**
 - General implementation of sample splitting

DoubleML: Model Overview

Model Type	Treatment	Data Structure	Causal Estimands
Partially Linear	Continuous / Binary	Cross-sectional	ATE, CATE, GATE
Interactive Regression	Binary	Cross-sectional	ATE, ATTE, CATE, GATE
Potential Outcome Models	Binary/Categorical	Cross-sectional	APO, CAPO, GAPO
Instrumental Variables	Continuous / Binary	Cross-sectional	ATE, LATE
Difference-in-Differences	Binary	Panel / Repeated CS	ATT
Sample Selection	Binary	Cross-sectional	ATE (selected)
Regression Discontinuity	Binary	Cross-sectional	LATE (at cutoff)

Extensions: Quantile treatment effects, conditional value at risk, weighted treatment effects, ...

DoubleML Workflow Example

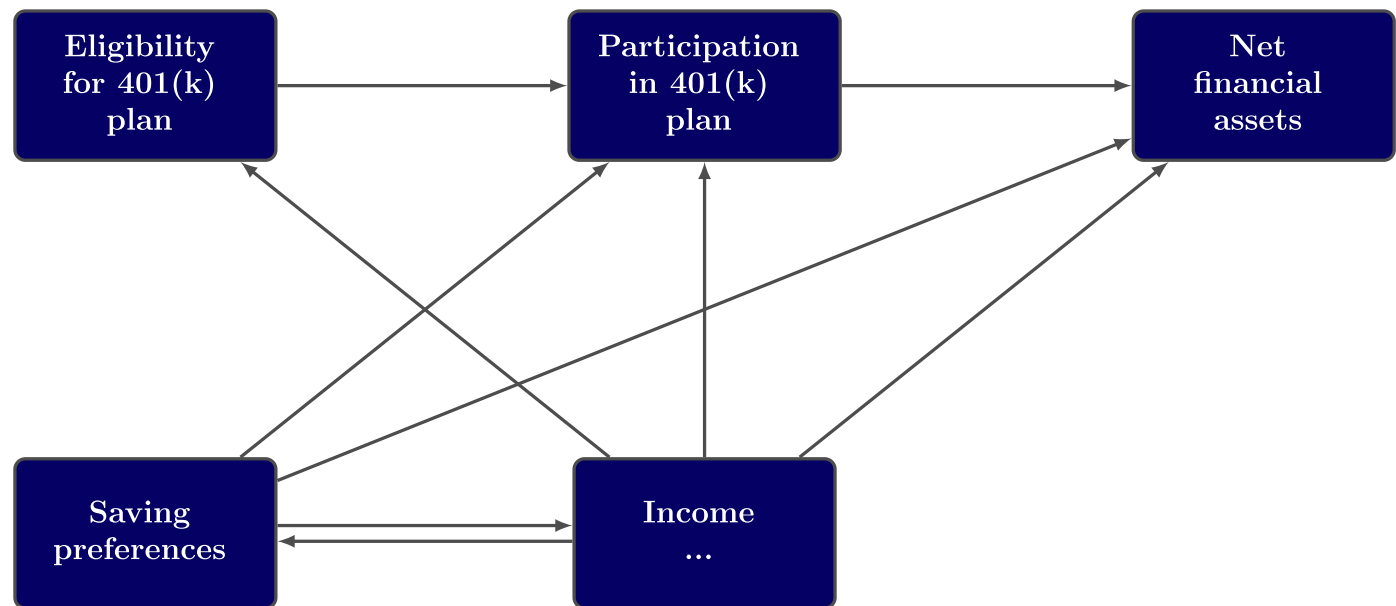
Workflow

o. Problem Formulation

1. Data-Backend
2. Causal Model
3. ML Methods
4. DML Specification
5. Hyperparameter Tuning
6. Estimation
7. Inference

o. Problem Formulation

- **401(k) Example**
- Goal: Estimate ATE of eligibility in 401(k) pension plans on employees' net financial assets



DAG, 401(k) example

DoubleML Workflow Example

Workflow

0. Problem Formulation

1. Data-Backend

- Declare the roles for the **treatment** variable, the **outcome** variable and **controls**

1. Data-Backend

2. Causal Model

3. ML Methods

4. DML Specification

5. Hyperparameter Tuning

6. Estimation

7. Inference

```
1 from doubleml.data import DoubleMLData
2 from doubleml.datasets import fetch_401K
3
4 data = fetch_401K(return_type='DataFrame')
5
6 # Construct DoubleMLData object
7 dml_data = DoubleMLData(
8     data,
9     y_col='net_tfa',
10    d_cols='e401',
11    x_cols=['age', 'inc', 'educ', 'fsize', 'marr',
12           'twoearn', 'db', 'pira', 'hown']
13 )
```

DoubleML Workflow Example

Workflow

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2. Causal Model

- Choose the causal model in DoubleML

Model Type	Treatment	Data Structure	Causal Estimands
Partially Linear	Continuous / Binary	Cross-sectional	ATE, CATE, GATE
Interactive Regression	Binary	Cross-sectional	ATE, ATTE, CATE, GATE
Potential Outcome Models	Binary/Categorical	Cross-sectional	APO, CAPO, GAPO
Instrumental Variables	Continuous / Binary	Cross-sectional	ATE, LATE
Difference-in-Differences	Binary	Panel / Repeated CS	ATT
Sample Selection	Binary	Cross-sectional	ATE (selected)
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DoubleML Workflow Example

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3. ML Methods

- Initialize the **learners** with hyperparameters
- Internal tuning is optional

Random forest

XGBoost

```
1 # Random forest learners
2 from sklearn.ensemble import RandomForestClassifier, RandomForestRegressor
3
4 ml_l_rf = RandomForestRegressor(
5     n_estimators = 500,
6     max_depth = 7,
7     max_features = 3,
8     min_samples_leaf = 3
9 )
10
11 ml_m_rf = RandomForestClassifier(
12     n_estimators = 500,
13     max_depth = 5,
14     max_features = 4,
15     min_samples_leaf = 7
16 )
```


DoubleML Workflow Example

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4. DML Specifications

- Initialize the model **DoubleML** object for the causal model, here **DoubleMLIRM**

Random forest

XGBoost

```
1 import numpy as np
2 import pandas as pd
3 from doubleml import DoubleMLIRM
4
5 np.random.seed(42)
6 # Default values
7 dml_irm_rf = DoubleMLIRM(
8     dml_data,
9     ml_g = ml_l_rf,
10    ml_m = ml_m_rf
11 )
```

DoubleML Workflow Example

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5. Hyperparameter Tuning

Random forest

XGBoost

► Hyperparameter spaces & Settings

```
1 # Run hyperparameter tuning
2 dml_irm_rf.tune_ml_models(
3     ml_param_space=param_space,
4     optuna_settings=optuna_settings,
5 )
```

Best trial: 2. Best value: -45821.2: 100%10/10 [00:07<00:00, 1.22it/s]

Best trial: 3. Best value: -62834.7: 100%10/10 [00:06<00:00, 1.48it/s]

Best trial: 5. Best value: -0.572394: 100%10/10 [00:18<00:00, 1.46s/it]

<doubleml.irm.irm.DoubleMLIRM at 0x7fadf6608860>

DoubleML Workflow Example

Workflow

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6. Estimation

- Use the `fit()` method to estimate the model

Random forest

XGBoost

```
1 # Estimation
2 dml_irm_rf.fit()
3
4 # Coefficient estimate
5 dml_irm_rf.coef
```

array([8135.69243711])

```
1 # Standard error
2 dml_irm_rf.se
```

array([1113.93596101])

```
1 # Summary
2 dml_irm_rf.summary.round(2)
```

	coef	std err	t	P> t	2.5 %	97.5 %
e401	8135.69	1113.94	7.3	0.0	5952.42	10318.97

DoubleML Workflow Example

Workflow

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7. Inference

6. Estimation

- For an overview on DoubleML objects use the `print()` method

Random forest

XGBoost

```
1 # Estimation
2 print(dml_irm_rf)
```

```
===== DoubleMLIRM Object =====

----- Data Summary -----
Outcome variable: net_tfa
Treatment variable(s): ['e401']
Covariates: ['age', 'inc', 'educ', 'fsize', 'marr', 'twoearn', 'db', 'pira',
'hown']
Instrument variable(s): None
No. Observations: 9915

----- Score & Algorithm -----
Score function: ATE

----- Machine Learner -----
Learner ml_g: RandomForestRegressor(max_depth=7, max_features=3,
min_samples_leaf=3,
                                n_estimators=500)
Learner ml_m: RandomForestClassifier(max_depth=5, max_features=4,
min_samples_leaf=7,
                                n_estimators=500)
DoubleML Training © 2020 Economic
```

DoubleML Workflow Example

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- 6. Estimation**
7. Inference

6. Estimation

- Additional diagnostics: Predictive performance of the ML learners

Random forest

XGBoost

```
1 # Estimation
2 print(dml_irm_rf.evaluate_learners())
```

```
{'ml_g0': array([[47704.16809874]]), 'ml_g1': array([[63905.21181478]]),
'ml_m': array([[0.44310813]])}
```

DoubleML Workflow Example

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7. Inference

- For confidence intervals use the `confint()` method

Random forest		XGBoost				
<pre>1 # Summary 2 dml_irm_rf.summary.round(2)</pre>						
	coef	std err	t	P> t	2.5 %	97.5 %
e401	8135.69	1113.94	7.3	0.0	5952.42	10318.97
<pre>1 # Confidence intervals 2 dml_irm_rf.confint(level=0.95).round(2)</pre>						
					2.5 %	97.5 %
e401					5952.42	10318.97

DoubleML Workflow Example

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7. Inference

- For confidence intervals use the `confint()` method

Random forest

XGBoost

```
1 # Multiplier bootstrap (relevant in case with multiple treatment variables)
2 _ = dml_irm_rf.bootstrap()
3
4 dml_irm_rf.confint(joint=True)
```

	2.5 %	97.5 %
e401	5985.440818	10285.944056

Full Example: 401(k) Data

A more detailed version of workflow example is available on

 docs.doubleml.org

and in the

 [DoubleML example gallery](#)

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Appendix

FAQs

Relation to other libraries for Causal ML

- **DoubleML** provides a general implementation of the Double Machine Learning approach by Chernozhukov et al. (2018) in Python and R
- There are also other open source libraries available for causal machine learning
 - **CausalML** (uber, <https://github.com/uber/causalml>, Chen et al. (2020)) - variety of causal ML learners, i.a. with focus on uplift modeling, CATEs and IATEs
 - **EconML** (microsoft research, <https://github.com/microsoft/EconML>, Microsoft Research (2019)) - various causal estimators based on machine learning, among others based on the double machine learning approach
 - ...

FAQs

Relation to other libraries for Causal ML

CausalML and **EconML** have a focus on heterogeneity of treatment effects from their start on

DoubleML focuses on implementing the DML approach and its extensions (example: heterogeneity, diff-in-diff, quantile regression, ...)

→ Object-orientated implementation based on orthogonal score

→ Extendibility and flexibility

References

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